Elias Stengel-Eskin

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Education

Johns Hopkins University — Baltimore, USA
Ph.D. in Computer Science – *Advisor: Benjamin Van Durme* — 2018–2023

- Ph.D. thesis: "Modeling Meaning for Description and Interaction"
- supported by NSF Graduate Research Fellowship

Johns Hopkins University — Baltimore, USA
MSE in Computer Science – *Advisor: Benjamin Van Durme* — 2018–2021

McGill University — Montréal, Canada
Bachelor of Arts and Sciences in Cognitive Science — 2014–2018

- Minor: Linguistics
- First Class Honours
- Honours thesis: "Variational Bayesian Inference for Unsupervised Lexicon Discovery" – *Advisor: Timothy O'Donnell*

Experience

The University of Texas at Austin
Assistant Professor, Department of Computer Science — August 2025–Present

University of North Carolina, Chapel Hill
Postdoctoral Research Associate – *Advisor: Mohit Bansal* — August 2023–August 2025

Microsoft Research
PhD Research Intern – *Advisors: Marc-Alexandre Côté, Eric Yuan, Pierre-Yves Oudeyer* — March 2022-March 2023

Microsoft Research - Semantic Machines
PhD Research Intern – *Advisor: Yu Su* — May 2021 - August 2021

Montreal Computational and Quantitative Linguistics Lab
Research Assistant – *Advisor: Morgan Sonderegger* — 2016-2018

Publications (peer-reviewed)

[1] Duy Nguyen, Hanqi Xiao, Archiki Prasad, Zaid Khan, Anirban Das, Austin Zhang, Sambit Sahu, Hyunji Lee, **Elias Stengel-Eskin**, and Mohit Bansal, "AVSD: Adaptive-view self-distillation by balancing consensus and teacher-specific privileged signals", *Advances in Neural Information Processing Systems 40*, 2026.

[2] Hyunji Lee, Justin Chih-Yao Chen, Joykirat Singh, Zaid Khan, **Elias Stengel-Eskin**, and Mohit Bansal, "LongMINT: Evaluating memory under multi-target interference in long-horizon agent systems", *Advances in Neural Information Processing Systems 40*, 2026.

[3] Eran Hirsch, David Wan, Han Wang, **Elias Stengel-Eskin**, Mohit Bansal, and Ido Dagan, "Who is the agent to blame? Localizing faithfulness and citation mistakes in agentic deep research", *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

[4] Zhengping Jiang, Gonzalo Ramos, Jina Suh, Shiqian Rachel Ng, **Elias Stengel-Eskin**, Justin Svegliato, Benjamin Van Durme, Andy Huntington, and Sam Thomson, "Frankenreport: Early exiting in long-form generation using expected value of computation", *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

[5] Emmanouil Zaranis, António Farinhas, Saul Santos, Beatriz Canaverde, Miguel Moura Ramos, Aditya K Surikuchi, André Viveiros, Baohao Liao, Elena Bueno-Benito, Nithin Sivakumaran, Pavlo Vasylenko, Shoubin Yu, Sonal Sannigrahi, Wafaa Mohammed, Ben Peters, Danae Sánchez Villegas, **Elias Stengel-Eskin**, *et al.*, "How well do vision-language models understand movies?", *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

[6] Atharv Kulkarni, Archiki Prasad, Vivek Srikumar, **Elias Stengel-Eskin**, Hyunji Lee, and Mohit Bansal, "Debug or regenerate? Test-guided selection for LLM code repair", *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

[7] Jihyung Park, Minchao Huang, Leqi Liu, and **Elias Stengel-Eskin**, "PragReST: Self-reinforcing counterfactual reasoning for pragmatic language understanding", *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

[8] Nithin Sivakumaran, Shoubin Yu, Hyunji Lee, Yue Zhang, Ali Payani, Mohit Bansal, and **Elias Stengel-Eskin**, "Balancing faithfulness and performance in reasoning via multi-listener soft execution", *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

[9] Jaewoo Lee, Zaid Khan, Archiki Prasad, Justin Chih-Yao Chen, Supriyo Chakraborty, Kartik Balasubramaniam, Sambit Sahu, **Elias Stengel-Eskin**, Hyunji Lee, and Mohit Bansal, "A history-aware visually grounded critic for computer use agents", *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

[10] Thomas Hofweber, Peter Hase, **Elias Stengel-Eskin**, and Mohit Bansal, "Are language models rational? The case of coherence norms and belief revision", *Artificial Intelligence*, 2026.

[11] Atin Pothiraj, Jaemin Cho, Yue Zhang, **Elias Stengel-Eskin**, and Mohit Bansal, "Physics question scene graph: Fine-grained evaluation of physical plausibility in text-to-video generation", *The 19th European Conference on Computer Vision (ECCV)*, 2026.

[12] Hanqi Xiao, Vaidehi Patil, Zaid Khan, Hyunji Lee, **Elias Stengel-Eskin**, and Mohit Bansal, "Playing along: Learning a double-agent defender for belief steering via theory of mind", *Conference on Language Modeling (COLM)*, 2026.

[13] Joykirat Singh, Justin Chih-Yao Chen, Archiki Prasad, **Elias Stengel-Eskin**, Akshay Nambi, and Mohit Bansal, "Think right: Learning to mitigate under-over thinking via adaptive, attentive compression", *Conference on Language Modeling (COLM)*, 2026.

[14] Xinyu Yang, Junlin Han, Rishi Bommasani, Jinqi Luo, Wenjie Qu, Wangchunshu Zhou, Adel Bibi, Xiyao Wang, Jaehong Yoon, **Elias Stengel-Eskin**, *et al.*, "Reliable and responsible foundation models: A comprehensive survey", *Transactions on Machine Learning Research (TMLR)*, 2026.

[15] Justin Chih-Yao Chen, Sukwon Yun, **Elias Stengel-Eskin**, Tianlong Chen, and Mohit Bansal, "Skill-based Mixture-of-Experts: Adaptive routing for heterogeneous reasoning via inferred skills", *The 43rd International Conference on Machine Learning (ICML)*, 2026.

[16] David Wan, Han Wang, Ziyang Wang, **Elias Stengel-Eskin**, Hyunji Lee, and Mohit Bansal, "Multimodal fact-level attribution for verifiable reasoning", *The 43rd International Conference on Machine Learning (ICML)*, 2026.

[17] Hanqi Xiao, Vaidehi Patil, Hyunji Lee, **Elias Stengel-Eskin**, and Mohit Bansal, "Generalized correctness models: Learning calibrated and model-agnostic correctness predictors from historical patterns", *The 43rd International Conference on Machine Learning (ICML)*, 2026.

[18] Tianyi Niu, Justin Chih-Yao Chen, Genta Indra Winata, Shi-Xiong Zhang, Supriyo Chakraborty, Sambit Sahu, Yue Zhang, **Elias Stengel-Eskin**, and Mohit Bansal, "Routing with generated data: Annotation-free LLM skill estimation and expert selection", *Proceedings of the Sixty-Fourth Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

[19] Jaewoo Lee, Archiki Prasad, Justin Chih-Yao Chen, Zaid Khan, **Elias Stengel-Eskin**, and Mohit Bansal, "PRInTS: Reward modeling for long-horizon information seeking", *Proceedings of the Sixty-Fourth Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

[20] Duy Nguyen, Archiki Prasad, **Elias Stengel-Eskin**, and Mohit Bansal, "GrAInS: Gradient-based attribution for inference-time steering of LLMs and VLMs", *Proceedings of the Sixty-Fourth Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

[21] Nithin Sivakumaran, Justin Chih-Yao Chen, David Wan, Yue Zhang, Jaehong Yoon, **Elias Stengel-Eskin**, and Mohit Bansal, "DART: Leveraging multi-agent disagreement for tool recruitment in multimodal reasoning", *Proceedings of the 2026 Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026.

[22] Joyce Horn Fonteles, Nithin Sivakumaran, Clayton Cohn, Austin Coursey, Shoubin Yu, **Elias Stengel-Eskin**, Ashwin TS, Mohit Bansal, and Gautam Biswas, "A novel approach to evaluating the effectiveness of large language models for multimodal analysis of embodied learning in classrooms", in *Proceedings of the LAK26: 16th International Learning Analytics and Knowledge Conference*, 2026, pages 536–546.

[23] Tianyi Niu, Jaemin Cho, **Elias Stengel-Eskin**, and Mohit Bansal, "RotBench: Evaluating multimodal large language models on identifying image rotation", *Proceedings of the 2026 Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026.

[24] Hyunji Lee, Seunghyun Yoon, Yunjae Won, Hanseok Oh, Geewook Kim, Trung Bui, Franck Dernoncourt, **Elias Stengel-Eskin**, Mohit Bansal, and Minjoon Seo, "Instruction tuning with and without context: Behavioral shifts and downstream impact", *Proceedings of the 2026 Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026.

[25] Victor Wang and **Elias Stengel-Eskin**, "Calibrating verbalized confidence with self-generated distractors", *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.

[26] Hyunji Lee, Minseon Kim, Chinmay Singh, Matheus Pereira, Atharv Sonwane, Isadora White, **Elias Stengel-Eskin**, Mohit Bansal, Zhengyan Shi, Alessandro Sordoni, *et al.*, "Gistify! Codebase-level understanding via runtime execution", *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.

[27] Amith Ananthram, **Elias Stengel-Eskin**, Lorena A Bradford, Julia Demarest, Adam Purvis, Keith Krut, Robert Stein, Rina Elster Pantalony, Mohit Bansal, and Kathleen McKeown, "PoSh: Using scene graphs to guide LLMs-as-a-judge for detailed image descriptions", *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.

[28] Zaid Khan, Archiki Prasad, **Elias Stengel-Eskin**, Jaemin Cho, and Mohit Bansal, "One life to learn: Inferring symbolic world models for stochastic environments from unguided exploration", *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.

[29] Yue Huang, Chujie Gao, Siyuan Wu, Haoran Wang, Xiangqi Wang, Jiayi Ye, Yujun Zhou, Yanbo Wang, Jiawen Shi, Qihui Zhang, Han Bao, Zhaoyi Liu, Yuan Li, Tianrui Guan, Peiran Wang, Haomin Zhuang, Dongping Chen, Kehan Guo, Andy Zou, Bryan Hooi, Caiming Xiong, **Elias Stengel-Eskin**, *et al.*, "TrustGen: A platform of dynamic benchmarking on the trustworthiness of generative foundation models", *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.

[30] Jio Choi, Mohit Bansal, and **Elias Stengel-Eskin**, "Language models identify ambiguities and exploit loopholes", *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

[31] Justin Chih-Yao Chen, Archiki Prasad, Swarnadeep Saha, **Elias Stengel-Eskin**, and Mohit Bansal, "Magicore: Multi-agent, iterative, coarse-to-fine refinement for reasoning", *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

[32] Duy Nguyen, Archiki Prasad, **Elias Stengel-Eskin**, and Mohit Bansal, "Laser: Learning to adaptively select reward models with multi-armed bandits", *Advances in Neural Information Processing Systems 39*, 2025.

[33] Eduardo Calò, Guanyi Chen, **Elias Stengel-Eskin**, Albert Gatt, and Kees van Deemter, "Incorporating formulaicness in the automatic evaluation of naturalness: A case study in logic-to-text generation", in *Proceedings of the 18th International Natural Language Generation Conference*, 2025, pages 352–365.

[34] Nithin Sivakumaran, Chia-Yu Yang, Abhay Zala, Shoubin Yu, Daeun Hong, Xiaotian Zou, **Elias Stengel-Eskin**, Dan Carpenter, Wookhee Min, Cindy Hmelo-Silver, *et al.*, "A multimodal classroom video question-answering framework for automated understanding of collaborative learning", in *Proceedings of the 27th International Conference on Multimodal Interaction*, 2025, pages 516–525.

[35] Han Wang, Archiki Prasad, **Elias Stengel-Eskin**, and Mohit Bansal, "Retrieval-augmented generation with conflicting evidence", *Conference on Language Modeling (COLM)*, 2025.

[36] Hanqi Xiao, Yi-Lin Sung, **Elias Stengel-Eskin**, and Mohit Bansal, "Task-circuit quantization: Leveraging knowledge localization and interpretability for compression", *Conference on Language Modeling (COLM)*, 2025.

[37] Archiki Prasad, **Elias Stengel-Eskin**, Justin Chih-Yao Chen, Zaid Khan, and Mohit Bansal, "Learning to generate unit tests for automated debugging", *Conference on Language Modeling (COLM)*, 2025.

[38] David Wan, Eran Hirsch, **Elias Stengel-Eskin**, Ido Dagan, and Mohit Bansal, "Generationprograms: Fine-grained attribution with executable programs", *Conference on Language Modeling (COLM)*, 2025.

[39] Atin Pothiraj, **Elias Stengel-Eskin**, Jaemin Cho, and Mohit Bansal, "Capture: Evaluating spatial reasoning in vision language models via occluded object counting", *Proceedings of the International Conference on Computer Vision, (ICCV)*, 2025.

[40] Duy Nguyen, Archiki Prasad, **Elias Stengel-Eskin**, and Mohit Bansal, "Multi-attribute steering of language models via targeted intervention", *Proceedings of the Sixty-Third Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[41] Eran Hirsch, Aviv Slobodkin, David Wan, **Elias Stengel-Eskin**, Mohit Bansal, and Ido Dagan, "Laquer: Localized attribution queries in content-grounded generation", *Proceedings of the Sixty-Third Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[42] Ziyang Wang*, Shoubin Yu*, **Elias Stengel-Eskin***, Jaehong Yoon, Feng Cheng, Gedas Bertasius, and Mohit Bansal, "VideoTree: Adaptive tree-based video representation for LLM reasoning on long videos", *Proceedings of the Forty-Second Annual IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[43] Zaid Khan, **Elias Stengel-Eskin**, Jaemin Cho, and Mohit Bansal, "DataEnvGym: Data generation agents in teacher environments with student feedback", *The Thirteenth International Conference on Learning Representations (ICLR)* ***Spotlight***, 2025.

[44] Swarnadeep Saha, Archiki Prasad, Justin Chih-Yao Chen, Peter Hase, **Elias Stengel-Eskin**, and Mohit Bansal, "System-1.x: Learning to balance fast and slow planning with language models", *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

[45] Amith Ananthram, **Elias Stengel-Eskin**, Carl Vondrick, Mohit Bansal, and Kathleen McKeown, "See it from my perspective: Diagnosing the western cultural bias of large vision-language models in image understanding", *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

[46] **Elias Stengel-Eskin**, Peter Hase, and Mohit Bansal, "Teaching models to balance resisting and accepting persuasion", *Proceedings of the Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.

[47] David Wan, Justin Chih-Yao Chen, **Elias Stengel-Eskin**, and Mohit Bansal, "MAMM-Refine: A recipe for improving faithfulness in generation with multi-agent collaboration", *Proceedings of the Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.

[48] Han Wang, Archiki Prasad, **Elias Stengel-Eskin**, and Mohit Bansal, "AdaCAD: Adaptively decoding to balance conflicts between contextual and parametric knowledge", *Proceedings of the Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.

[49] **Elias Stengel-Eskin**, Peter Hase, and Mohit Bansal, "LACIE: Listener-aware finetuning for confidence calibration in large language models", *Advances in Neural Information Processing Systems 38*, 2024.

[50] Jinhao Duan, Renming Zhang, James Diffenderfer, Bhavya Kailkhura, Lichao Sun, **Elias Stengel-Eskin**, Mohit Bansal, Tianlong Chen, and Kaidi Xu, "GTBench: Uncovering the strategic reasoning limitations of LLMs via game-theoretic evaluations", *Advances in Neural Information Processing Systems 38*, 2024.

[51] Ansel Blume, Khanh Duy Nguyen, Zhenhailong Wang, Yangyi Chen, Michal Shlapentokh-Rothman, Xiaomeng Jin, Jeonghwan Kim, ..., **Stengel-Eskin-Elias**, *et al.*, "MIRACLE: An online, explainable multimodal interactive concept learning system", in *Proceedings of the 32nd ACM International Conference on Multimedia*, 2024.

[52] Peter Hase, Thomas Hofweber, Xiang Zhou, **Elias Stengel-Eskin**, and Mohit Bansal, "Fundamental problems with model editing: How should rational belief revision work in LLMs?", *Transactions on Machine Learning*, 2024.

[53] **Elias Stengel-Eskin**, Archiki Prasad, and Mohit Bansal, "ReGAL: Refactoring programs to discover generalizable abstractions", *The 41st International Conference on Machine Learning (ICML)*, 2024.

[54] Justin Chih-Yao Chen, Swarnadeep Saha, **Elias Stengel-Eskin**, and Mohit Bansal, "MAGDi: Structured distillation of multi-agent interaction graphs improves reasoning in smaller language models", *The 41st International Conference on Machine Learning (ICML)*, 2024.

[55] Haotian Fu, Pratyusha Sharma, **Elias Stengel-Eskin**, George Konidaris, Nicolas Le Roux, Marc-Alexandre Côté, and Xingdi Yuan, "Language-guided skill learning with temporal variational inference", *The 41st International Conference on Machine Learning (ICML)*, 2024.

[56] David Wan, Jaemin Cho, **Elias Stengel-Eskin**, and Mohit Bansal, "Contrastive region guidance: Improving grounding in vision-language models without training", *The 18th European Conference on Computer Vision (ECCV)*, 2024.

[57] Han Wang*, Archiki Prasad*, **Elias Stengel-Eskin***, and Mohit Bansal, "Soft self-consistency improves language model agents", *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[58] Maryam Hashemzadeh, **Elias Stengel-Eskin**, Sarath Chandar, and Marc-Alexandre Cote, "Sub-goal distillation: A method to improve small language agents", *The 3rd Conference on Lifelong Learning Agents (CoLLAs)*, 2024.

[59] **Elias Stengel-Eskin**, Kyle Rawlins, and Benjamin Van Durme, "Zero and few-shot semantic parsing with ambiguous inputs", in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[60] Archiki Prasad, **Elias Stengel-Eskin**, and Mohit Bansal, "Rephrase, augment, reason: Visual grounding of questions for vision-language models", in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[61] **Elias Stengel-Eskin** and Benjamin Van Durme, "Did You Mean...? Confidence-based Trade-offs in Semantic Parsing", *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[62] **Elias Stengel-Eskin** and Benjamin Van Durme, "Calibrated Interpretation: Confidence Estimation in Semantic parsing", *Transactions of the Association for Computational Linguistics (TACL)*, 2023.

[63] **Elias Stengel-Eskin**, Jimena Guallar-Blasco, Yi Zhou, and Benjamin Van Durme, "Why Did the Chicken Cross the Road? Rephrasing and Analyzing Ambiguous Questions in VQA", *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

[64] Zhuowan Li, Xingrui Wang, **Elias Stengel-Eskin**, Adam Kortylewski, Wufei Ma, Benjamin Van Durme, and Alan Yuille, "Super-CLEVR: A Virtual Benchmark to Diagnose Domain Robustness in Visual Reasoning", *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[65] **Elias Stengel-Eskin** and Benjamin Van Durme, "The Curious Case of Control", *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.

[66] **Elias Stengel-Eskin**, Emmanouil Antonios Platanios, Adam Pauls, Sam Thomson, Hao Fang, Benjamin Van Durme, Jason Eisner, and Yu Su, "When More Data Hurts: A Troubling Quirk in Developing Broad-Coverage Natural Language Understanding Systems", *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.

[67] Chenyu Zhang, Benjamin Van Durme, Zhuowan Li, and **Elias Stengel-Eskin**, "Visual Commonsense in Pretrained Unimodal and Multimodal Models", in *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, Seattle, Washington: Association for Computational Linguistics, 2022.

[68] **Elias Stengel-Eskin**, Andrew Hundt, Zhuohong He, Aditya Murali, Nakul Gopalan, Matthew Gombolay, and Gregory D. Hager, "Guiding Multi-Step Rearrangement Tasks with Natural Language Instructions", in *5th Annual Conference on Robot Learning (CoRL)*, 2021.

[69] Zhuowan Li, **Elias Stengel-Eskin**, Yixiao Zhang, Cihang Xie, Quan Tran, Benjamin Van Durme, and Alan Yuille, "Calibrating Concepts and Operations: Towards Symbolic Reasoning on Real Images", in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[70] **Elias Stengel-Eskin**, Jimena Guallar-Blasco, and Benjamin Van Durme, "Human-Model Divergence in the Handling of Vagueness", in *Proceedings of the 1st Workshop on Understanding Implicit and Underspecified Language (UnImplicit)*, Online: Association for Computational Linguistics, 2021.

[71] **Elias Stengel-Eskin**, Kenton Murray, Sheng Zhang, Aaron Steven White, and Benjamin Van Durme, "Joint Universal Syntactic and Semantic Parsing", *Transactions of the Association for Computational Linguistics (TACL)*, 2021.

[72] **Elias Stengel-Eskin**, Jimena Guallar-Blasco, and Benjamin Van Durme, "Exploring Human-Model Divergence Through Vagueness", *Proceedings of the Society for Computation in Linguistics (SCiL)*, 2021, ***Abstract**.

[73] Ryan Culkin, J. Edward Hu, **Elias Stengel-Eskin**, Guanghui Qin, and Benjamin Van Durme, "Iterative Paraphrastic Augmentation with Discriminative Span Alignment", *Transactions of the Association for Computational Linguistics (TACL)*, volume 9, 2021.

[74] **Elias Stengel-Eskin**, Aaron Steven White, Sheng Zhang, and Benjamin Van Durme, "Universal Decompositional Semantic Parsing", in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

[75] Aaron Steven White, **Elias Stengel-Eskin**, Siddharth Vashishtha, Venkata Subrahmanyan Govindarajan, Dee Ann Reisinger, Tim Vieira, Keisuke Sakaguchi, Sheng Zhang, Francis Ferraro, Rachel Rudinger, *et al.*, "The Universal Decompositional Semantics Dataset and Decomp Toolkit", in *Proceedings of The 12th Language Resources and Evaluation Conference*, 2020.

[76] **Elias Stengel-Eskin**, Tzu-Ray Su, Matt Post, and Benjamin Van Durme, "A Discriminative Neural Model for Cross-Lingual Word Alignment", in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019.

[77] Michael McAuliffe, **Elias Stengel-Eskin**, Michaela Socolof, and Morgan Sonderegger, "Polyglot and Speech Corpus Tools: A System for Representing, Integrating, and Querying Speech Corpora." In *INTERSPEECH*, 2017.

Publications (other)

1. Nithin Sivakumaran, Ali Payani, Hyunji Lee, Mohit Bansal, **Elias Stengel-Eskin**, "ToM2Vec: Continuous Theory-of-Mind Tokens for Multi-Agent Coordination" (2026), *Under Review*
2. Yanshu Li, Ruixiang Tang, Junyi Jessy Li, **Elias Stengel-Eskin**, "ActICL: Post-Training for Agentic Multimodal In-Context Learning" (2026), *Under Review*
3. Hyunji Lee, Joykirat Singh, Zaid Khan, Justin Chih-Yao Chen, **Elias Stengel-Eskin**, Alessandro Sordoni, Arman Cohan, Mohit Bansal, "Balancing Memory Pathways: Analyzing and Improving Memory Utilization in Hybrid LMs" (2026), *Under Review*
4. Justin Chih-Yao Chen, **Elias Stengel-Eskin**, Yan Chen, Pol Llado, Scott Counts, Mohit Bansal, Benjamin Van Durme, Harsh Jhamtani, Gaurav Verma, "TeleTune: Evolving Agent Skills From Offline Telemetry" (2026), *Under Review*
5. Victor Wang, Thomas Hofweber, Mohit Bansal, **Elias Stengel-Eskin**, "Evaluating Persistent Calibration under Evolving Model Knowledge" (2026), *Under Review*
6. **Elias Stengel-Eskin**, Newton Sander, Carlos Bonetti, Sasha Boguraev, James Bowler, Hale Sirin, Simon Kirby, "GlossoGen: Emergent Language in Complex Multi-Agent LLM Interactions" (2026), *Under Review*
7. Ananya Sahu, Mohit Bansal, **Elias Stengel-Eskin**, "CreativeInstruct: Scalably Teaching LLMs to Balance Quality, Creativity, and Diversity" (2026), *Under Review*
8. Ashwin Vinod, Ying Ding, **Elias Stengel-Eskin**, "CalVerT: Augmenting Agents with Calibrated Verifier Telemetry Improves Action and Learning in Knowledge-Intensive Tasks" (2026), *Under Review*
9. Zaid Khan, Justin Chih-Yao Chen, Jaemin Cho, **Elias Stengel-Eskin**, Mohit Bansal, "GPU Forecasters: Language Models as Selective Surrogates for Kernel Runtime Optimization" (2026), *Under Review*
10. Joykirat Singh, Zaid Khan, Archiki Prasad, Justin Chih-Yao Chen, Akshay Nambi, Hyunji Lee, **Elias Stengel-Eskin**, Mohit Bansal, "Agent-BRACE: Decoupling Beliefs from Actions in Long-Horizon Tasks via Verbalized State Uncertainty" (2026) *Under Review*
11. Saeid Asgari Taghanaki, Rakshanda Agarwal, Bruce Sun, Rohan Jha, **Elias Stengel-Eskin**, Sara Malvar, Rui Ying, Yifei Xu, Guilherme Potje, Tusher Chakraborty, Leonardo de Oliveira Nunes, Ranveer Chandra, Emre Kiciman, "Diagnosing Capability Gaps in Fine-Tuning Data" (2026) *Under Review*
12. Han Wang, David Wan, Hyunji Lee, Thinh Pham, Mikaela Cankosyan, Weiyuan Chen, **Elias Stengel-Eskin**, Tu Vu, Mohit Bansal, "MERRIN: A Benchmark for Multimodal Evidence Retrieval and Reasoning in Noisy Web Environments" (2026), *Under Review*

13. Justin Chih-Yao Chen, Archiki Prasad, Zaid Khan, Joykirat Singh, Runchu Tian, **Elias Stengel-Eskin**, Mohit Bansal, "Cog-DRIFT: Exploration on Adaptively Reformulated Instances Enables Learning from Hard Reasoning Problems" (2026), *Under Review*

14. Marvin Limpijankit, Milad Alshomary, Yassin Oulad Daoud, Amith Ananthram, Tim Trombley, **Elias Stengel-Eskin**, Mohit Bansal, Noam M. Elcott, Kathleen McKeown, "Does AI See like Art Historians? Interpreting How Vision Language Models Recognize Artistic Style" (2026), *Under Review*

15. Vaidehi Patil, **Elias Stengel-Eskin**, Mohit Bansal, "The Sum Leaks More Than Its Parts: Compositional Privacy Risks and Mitigations in Multi-Agent Collaboration" (2025), *Under Review*

16. Duy Nguyen, Hanqi Xiao, Archiki Prasad, **Elias Stengel-Eskin**, Hyunji Lee, Mohit Bansal, "Conflict-Resolving and Sharpness-Aware Minimization for Generalized Knowledge Editing with Multiple Updates" (2026), *Under Review*

17. Haotian Zhai, **Elias Stengel-Eskin**, Pratik Patil, Liu Leqi, "Evaluating Stochasticity in Deep Research Agents" (2026), *Under Review*

18. David Wan, Han Wang, **Elias Stengel-Eskin**, Jaemin Cho, Mohit Bansal, "CLaMR: Contextualized Late-Interaction for Multimodal Content Retrieval" (2025), *Under Review*

19. Ron Eliav, Arie Cattan, Eran Hirsch, Shahaf Bassan, **Elias Stengel-Eskin**, Mohit Bansal, Ido Dagan, "CLATTER: Comprehensive Entailment Reasoning for Hallucination Detection" (2025), *Under Review*

20. Zaid Khan, **Elias Stengel-Eskin**, Archiki Prasad, Jaemin Cho, Mohit Bansal, "Executable Functional Abstractions: Inferring Generative Programs for Advanced Math Problems" (2025), *Under Review*

21. Vaidehi Patil, **Elias Stengel-Eskin**, Mohit Bansal, "UPCORE: Utility-Preserving Coreset Selection for Balanced Unlearning" (2025), *Under Review*

22. **Elias Stengel-Eskin**, "Modeling Meaning for Description and Interaction", (2023) **Ph.D. Thesis**

23. Jimena Guallar-Blasco, **Elias Stengel-Eskin**, Benjamin Van Durme, "Analyzing Question Ambiguity in *Why*-Questions" (2023), *Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL 2023)* *__Abstract__

24. Shalaka Vaidya, **Elias Stengel-Eskin**, Joao Sedoc, "Automatic Evaluation of Chit-chat via Semantic Parsing" (2022), *Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL 2022)* *__Abstract__

25. Yunmo Chen, Seth Ebner, Tongfei Chen, Patrick Xia, **Elias Stengel-Eskin**, Tzu-Ray Su, J. Edward Hu, Nils Holzenberger, Ryan Culkin, Craig Harman, Max Thomas, Thomas Lippincott, Aaron Steven White, Kyle Rawlins, Benjamin Van Durme, "NIST TAC SM-KBP 2019 System Description: JHU/UR Framework", (2019)

26. **Elias Stengel-Eskin**, "Variational Bayesian Inference for Unsupervised Lexicon Discovery" (2017), **Undergraduate Honours Thesis**

Funding

- **Cisco Faculty Award** 2026-2027
  $75k. Role: **PI**
- **Coefficient Giving: Meta Reasoning for AI Trustworthiness** 2026-2028
  $1,438,263. Role: **Co-PI** (PI: Kyle Mahowald; Co-PIs: Harvey Lederman, Jennifer Hu)
- **IARPA BENGAL** 2025-2027
  $290k. Role: Postdoc
- **NSF NAIRR Pilot Grant** 2024-2025
  Supervising Calibration for Safe and Trustworthy Artificial Intelligence Systems
  2,500 GPU Hours. Role: **PI**
- **Cisco Faculty Award** 2025

Improving LLM Reliability and Trustworthiness via Overconfidence Reduction and Conversation
$75K. Role: Co-PI

- **CapitalOne Gift** 2025
  Multi-Agent Systems and Mixtures-of-Experts for Robust Generation
  Role: Co-PI
- **Center for AI Safety Compute Grant** 2024
  Listener-Aware Confidence Calibration for Improving LLM Reliability and Trustworthiness
  Unlimited A100 GPU access. Role: Co-PI
- **NSF Graduate Research Fellowship** 2018-2023
  Role: Student

News Coverage/Appearances

1. *The AI That Cheated and Why Experts Are Concerned*, CBS Austin (KEYE), 07/31/2026
2. *OpenAI says AI acted on its own in an 'unprecedented' hack of another company*, KXAN News, 07/22/2026
3. *The importance of ambiguity*, Johns Hopkins University CS News, 04/15/2024
4. *Speaking easy: A simple method to clarify voice commands*, Johns Hopkins University CS News, 01/08/2024

Invited Talks/Panels

- **The Emergence of Agent Languages** November 2026
  Swarm Alignment Hackathon (Apart Research & AE Studio)
- **Multi-Model Training for Multi-Agent Communication Skills** 2026
  Keynote *– ICLR 2026 Workshop on Multi-Agent Learning in Generative AI*
- **Multi-Model Training for Multi-Agent Communication Skills** 2026
  Harrington Annual Symposium
- **Multi-Model Training for Multi-Agent Communication Skills** June 2, 2026
  Mila – STAMINA Working Group
- **Multi-Model Training for Multi-Agent Communication Skills** April 7, 2026
  Johns Hopkins University, DSAI Seminar
- **Multi-Model Training for Multi-Agent Communication Skills** April 6, 2026
  University of Pennsylvania, CLunch
- **Multi-Model Training for Multi-Agent Communication Skills** March 13, 2026
  Workshop on Large Language Models, Rice University
- **Panel**: Reimagining Work in the Age of AI Agents March 4, 2026
  Texas Symposium on Machine Learning, Responsible AI & Robotics
- **Towards Granular, Robust, Calibrated Judgments via Skill-Based and Persuasion-Balanced Reasoning** December 2, 2025
  NeurIPS Expo Workshop on Exploring Trust and Reliability in LLM Evaluation
- **Panel:** Agentic AI in Wireless Networking and Communications November 14, 2025
  UT Austin Wireless Networking and Communications Group
- **Panel:** The true impact of AI on the labor market September 30, 2025
  Hire EZ
- **Communication, Calibration, and Grounding for Collaborative AI Agents** January 8, 2025
  Duke University
- **Learning Pragmatic Skills through Multi-Agent Training** 2024

UNC Chapel Hill – Linguistics Department

- **Confidence-based Rephrasing, Refinement, and Selection** March 22nd, 2024
  Keynote *– UncertaiNLP Workshop, EACL 2024*
- **Ambiguity in NLP** October 26th, 2023
  UNC Chapel Hill – Linguistics Department
- **Uncertainty and Ambiguity in Semantic Structures** October 6th, 2023
  Emory University
- **Language, Structure, and Beyond** May 9th, 2023
  UNC Chapel Hill – MURGe Group
- **Language, Structure, and Beyond** April 4th, 2023
  Scaled Cognition
- **Language, Structure, and Beyond** March 13th, 2023
  Brandeis University – CS Department
- **Language, Structure, and Beyond** March 2nd, 2023
  University of Cambridge – CS Department
- **Language, Structure, and Beyond** February 17th, 2023
  George Mason University – George Mason NLP Group
- **Language, Structure, and Beyond** February 1st, 2023
  Georgetown University – NERT Lab
- **Joint Universal Syntactic and Semantic Parsing** April 1st, 2022
  Cornell University – Workshop on Meaning in Language

Honors and Awards

Honors and Awards

- First Class Honours in Cognitive Science 2018
- Dean's Honor List (top 10% of faculty) 2014–2015, 2016–2018
- Arts Undergraduate Research Internship Award 2016
- NSF Graduate Research Fellowship 2019

Service

Reviewing

- **Senior Area Chair:** EMNLP 2026 (Outstanding SAC Award), NAACL 2027
- **Area Chair:** ARR (2024: 2x, 2025: 4x, 2026: 3x), COLM (2026), NeurIPS (2026)
- **Reviewer:** ACL (2021, 2022), NAACL 2022, EACL 2023, CoRL 2023, EMNLP 2023, NeurIPS (2024, 2025), WiNLP (2024, 2025, 2026), ICLR (2025, 2026), ICML (2025, 2026 – Gold Reviewer), ICCV 2025, TACL (2025, 2026), CVPR 2026, ECCV 2026, ARR 2026, ICLR 2027
- **Program Committee**: UnImplicit 2022, SouthNLP 2024
- **Grant Reviewer:** Schmidt Sciences (AI2050 Fellowships 2026; Interpretability 2026; Multi-Agent Systems 2026)

Organizing

- **Workshop Organizer:** UnImplicit 2024
- **Conference Organizer:** EMNLP 2024 (Publicity Chair), ACL 2026 (Student Research Workshop Chair)
- **Session Chair:** NAACL 2025

PhD/Proposal/RPE Committees at UT Austin

- **Ph.D. Thesis Committees:** Liyan Tang, Vanya Cohen, Caroline Wang,
- **PhD Proposals:** Anibal Solon Heinsfeld, Vanya Cohen, Liyan Tang, Caroline Wang, Anuj Diwan, Shenghui Chen, Yunhao Yang, Lunyiu Nie
- **RPEs:** Fangcong Yin, Meng Chen, Parikshit Bansal, Ananya Gubbi Mohanbabu, Wei-Cheng Tseng, Daniel Young
- **M.S. Thesis Committees:** Emily Stewart
- **Undergraduate Thesis Committees:** Jocelyn Zhang, Willow Stenglein

External Committees

- **PhD Proposals:** Amith Ananthram (Columbia)

Department Service

- **UT Austin Graduate Admissions Committee**, 2025-2026
- **UT Austin Linguistics Hiring Committee**, 2025-2026
- **CLSP Visit Weekend Committee**, 2018-2022
- **CLSP Admissions Committee**, 2022-2023
- **Application Mentoring Group**, 2022-2023

Teaching

- **Instructor** at University of Texas at Austin — Fall 2025
  Communicative and Grounded AI Agents – CS395T
- **Guest Lecturer** at UNC Chapel Hill — Fall 2024
  Advanced Topics in Natural Language Processing (Aligning LLMs and Humans – COMP 790-158)
- **Guest Lecturer** at UNC Chapel Hill — Fall 2024
  Deep Learning (COMP 664)
- **Guest Lecturer** at UNC Chapel Hill — Fall 2023, Fall 2024
  Computational Linguistics
- **Teaching Assistant** at Johns Hopkins University — Fall 2019
  Artificial Intelligence (EN.601.464/664)

Mentoring

- Vaidehi Patil — Fall 2024-Present
  Ph.D. Student (UNC)
- Zaid Khan — Fall 2024-Present
  Ph.D. Student (UNC)
- Duy Nguyen — Fall 2024-Present
  Ph.D. Student (UNC)
- Amith Ananthram — Winter 2024-Present
  Ph.D. Student (Columbia)
- Ziyang Wang — Fall 2023-Fall 2024
  Ph.D. Student (UNC)
- Shoubin Yu — Fall 2023-Present
  Ph.D. Student (UNC)

- David Wan — Fall 2023-Present
  Ph.D. Student (UNC)
- Justin Chih-Yao Chen — Fall 2023-Present
  Ph.D. Student (UNC)
- Han Wang — Fall 2023-Present
  Ph.D. Student (UNC)
- Archiki Prasad — Summer 2023-Present
  Ph.D. Student (UNC)
- Victor Wang — Summer 2025-Present
  Undergrad (UT Austin)
  Outstanding Undergraduate Researcher Award**, CRA (Honorable Mention), **NSF GRFP
- Tianyi Niu — Summer 2025-Present
  MS (UNC)
- Hanqi Xiao — Summer 2024-Present
  Undergrad (UNC, Current: NYU PhD student)
  Outstanding Undergraduate Researcher Award**, CRA (Honorable Mention), **NSF GRFP
- Atin Pothviraj — Summer 2024-Present
  Undergrad (UNC)
- Nithin Sivakumaran — Summer 2024-Present
  Undergrad (UNC, Current: UMD PhD student)
  Outstanding Undergraduate Researcher Award**, CRA (Honorable Mention), **NSF GRFP
- Bryan Sukidi — Fall 2024-Fall 2025
  Undergrad (UNC)
- Evan Paces-Wiles — Fall 2024-Present
  Undergrad (UNC)
- Jio Choi — Fall 2024-Summer 2025
  Undergrad (UNC)
- Jimena Guallar-Blasco — Summer 2020-Summer 2024
  Undergrad/MS 2024 (JHU, Current: Charles River Analytics)
  ***Outstanding Undergraduate Researcher Award**, CRA (Honorable Mention)*
- Maryam Hashemzadeh — Summer 2023
  Ph.D. Student (MILA)
- Zhuohong (Zooey) He — Winter 2021-Spring 2021
  MSE, Spring 2021 (JHU, Current: Intuitive)
- Chenyu (Heidi) Zhang — Fall 2021-Fall 2022
  BS 2022, (JHU, Next: MS (Stanford), Current: Google Gemini)
- Shalaka Vaidya — Fall 2022
  MS 2023 (NYU)
- Yi Zhou — Winter 2022-Fall 2022
  MS 2022 (JHU)
- Vicky Zeng — Winter 2021-Winter 2022
  Ph.D. Student (JHU)

Skills

- **Programming (expert):** Python
- **Programming (proficient):** Bash, Java, Javascript
- **Libraries/Frameworks:** PyTorch, AllenNLP, Transformers, NLTK, numpy, MechanicalTurk, networkx, MXNet, React

Languages

- **Native:** English, German
- **Fluent:** French
- **Other:** Latin (reading/translation), Spanish (intermediate)